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NOETHER'S THEOREM AND THE CASIMIR HONEYCOMB DRIVE: ON NON-CONSERVATION OF MOMENTUM IN ASYMMETRIC VACUUM SYSTEMS

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The Casimir honeycomb drive [1] — a perfectly conducting plate with a nanoscale honeycomb structure on one side — is examined from the standpoint of symmetries. Because the vacuum modes on the two sides of the plate differ, the system is not invariant under translations along the normal: the vacuum Hamiltonian depends explicitly on the coordinate l, and by Noether's theorem the field momentum along this axis is not conserved. The thrust equals $\partial\langle H\rangle/\partial l$. The same force is then computed independently, from the Maxwell stress tensor built out of the fields of the cavity eigenmodes. Mode by mode the two derivations coincide identically (Hellmann–Feynman theorem); after regularization they coincide in the limit $ak_m \to \infty$. Along the way an error in Appendix B of [1] is found and corrected: for TM modes the ratio of pressure to energy density is k_z^2/k^2, not unity. The stress-tensor calculation shows where the force acts: it is the jump of $\langle T_{zz}\rangle$ at the metal surface, which is why the momentum flux through any closed vacuum surface enclosing the plate is independent of the position of its faces. Finally, it is shown that the standard momentum balance (Tamm, § 105) with $\langle \mathbf{S}\rangle = 0$ loses exactly these jump terms. What is not conserved is the momentum of the field (and of the plate); the total momentum of the system plate + field + support is conserved.

Keywords: *Noether's theorem, Casimir effect, quantum vacuum, momentum non-conservation, Maxwell stress tensor, nanohoneycomb, Casimir thrust*

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INTRODUCTION

In [1] it was shown that a perfectly conducting plate, one side of which is smooth and the other covered with honeycomb cells of square cross-section $a \times a$, experiences a net force exerted by the electromagnetic vacuum. The force was computed as the difference of the zero-point energy densities on the two sides of the plate, regularized by means of the Euler–Maclaurin formula; for a cell size $a \approx 2\ \mu$m it turned out to be of the order of several dynes per square metre.

Such a result inevitably raises a question: how is a one-body force compatible with momentum conservation? In the classical two-plate Casimir configuration the forces on the plates are equal and opposite, and the question does not arise. Here there is no second body.

The present paper answers this question in three steps.

The first step is symmetry. Noether's theorem [2] links momentum conservation to translational invariance. The honeycomb plate breaks this invariance, so the field momentum need not be conserved, and the force equals the derivative of the vacuum Hamiltonian with respect to the plate position. This result coincides with the force of [1].

The second step is an independent check. If the force is real, the Maxwell stress tensor must give it as well. We build the tensor directly from the fields of the cavity eigenmodes and show that for every mode the stress-tensor and Hamiltonian derivations coincide. This check revealed an error in Appendix B of [1], which we correct; it does not affect the Hamiltonian result.

The third step is where the force acts and where the momentum goes. The stress-tensor calculation shows that the force is concentrated in the jump of the stress tensor at the metal surface. The standard momentum balance equation (Tamm, § 105 [4]) contains no such jumps; with a vanishing mean Poynting vector this is what creates the apparent paradox. We point out the missing term and state precisely which momentum is not conserved.

All numerical statements of the paper are verified in the accompanying Jupyter notebook [9].

STATEMENT OF THE PROBLEM

A plate of zero thickness lies in the plane $z = 0$ inside a cube L^3 with perfectly conducting walls (Fig. 1). On the side $z > 0$ the plate carries perfectly conducting cells of cross-section $a \times a$ whose walls reach the cube face $z = l$ (in [1] $l = L/2$). On the side $z < 0$ there is the smooth face of the plate and a large cavity $L \times L \times (L - l)$.

Each cavity is a rectangular resonator $a \times b \times h$: for a cell $a = b$, $h = l$; for the large cavity $a = b = L$, $h = L - l$. The wave numbers are $k_x = \pi n_x/a$, $k_y = \pi n_y/b$, $k_z = \pi n_z/h$; $K^2 = k_x^2 + k_y^2$, $k^2 = K^2 + k_z^2$, $\omega = ck$. The notation $\sum_{n=(0)1}^{\infty}$ means that terms with $n = 0$ enter with weight $\frac{1}{2}$: each set (n_x, n_y, n_z) corresponds to two polarizations, but with one zero index only to one [1].

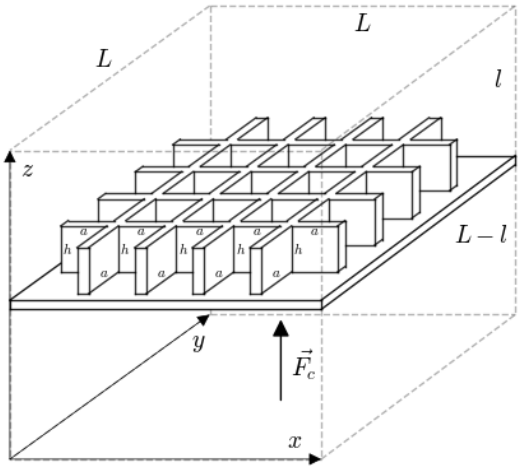


Fig.1. Cubic cavity with a plate covered by honeycomb.

The perfectly conducting walls of the cube, as in Casimir's work, are an auxiliary boundary condition without a physical counterpart, similar to the periodic boundary conditions used in quantum mechanics textbooks to normalize plane waves. The procedure is legitimate because the result is taken in the limit $L \to \infty$: by Weyl's theorem [7] the leading term of the mode count, $N(k) \simeq Vk^3/3\pi^2$, does not depend on the boundary conditions on the outer faces, only surface corrections do, and the contribution of the remote walls to the force on the plate vanishes. The cutoff k_m is a separate, physical element: it is set by the transparency of the plate material to short waves.

NOETHER'S THEOREM AND NON-CONSERVATION OF MOMENTUM

The sum of zero-point energies $E = \frac{1}{2}\sum\hbar\omega$ over both cavities gives the vacuum Hamiltonian per unit area [1]:

$$\frac{\langle 0|\hat{\mathscr{H}}|0\rangle}{S} = \frac{\hbar c}{a^2\pi}\left\{ l \sum_{n_x=(0)1}^{\infty}\sum_{n_y=(0)1}^{\infty}\int_0^\infty \sqrt{\frac{\pi^2}{a^2}(n_x^2+n_y^2)+k_z^2}\,dk_z + (L-l)\int_0^\infty\int_0^\infty\int_0^\infty \sqrt{k_x^2+k_y^2+k_z^2}\,dk_z\,\frac{a}{\pi}dk_x\,\frac{a}{\pi}dk_y \right\}.$$

This expression depends explicitly on l. Hence the system is **not invariant** under infinitesimal translations of the plate $z \to z + \delta z$.

By Noether's theorem the generator of spatial translations, the momentum component P_z, is conserved **if and only if** the Hamiltonian does not depend on z. Since $\partial\langle H\rangle/\partial l \neq 0$, the symmetry is broken, and P_z of the field is not conserved. The net force is

$$\frac{F}{S}=\frac{\partial}{\partial l}\frac{\langle 0|\hat{\mathscr{H}}|0\rangle}{S}=\frac{\hbar c}{a^2\pi}\left\{\sum_{n_x=(0)1}^{\infty}\sum_{n_y=(0)1}^{\infty}\int_0^\infty\sqrt{\frac{\pi^2}{a^2}(n_x^2+n_y^2)+k_z^2}\,dk_z-\int_0^\infty\int_0^\infty\int_0^\infty\sqrt{k_x^2+k_y^2+k_z^2}\,dk_z\,dn_x dn_y\right\},$$

which coincides with the thrust formula obtained in [1] from the difference of energy densities. After regularization by $f(k/k_m)$ and two-dimensional Euler–Maclaurin summation it takes the form $F/S=\delta(E/V)=R(ak_m)\hbar c\,\pi/a^4$.

Thus the appearance of thrust is not unexpected: it is a direct consequence of the broken translational symmetry. The quantum vacuum "presses" harder on one side because the mode spectra on the two sides of the plate differ. Let us state at once which momentum is meant: it is the translational symmetry of the boundary conditions, not of space, that is broken. Therefore it is the momentum of the field — or, equivalently, of the plate that absorbs it — that is not conserved; the total momentum of the system plate + field + support is conserved, just as for an ordinary sail in the wind. Where exactly the field transfers momentum to the plate will be shown by the stress tensor.

STRESS TENSOR FROM THE MODE FIELDS

Let us compute the force independently, through the Maxwell stress tensor, without assuming in advance the momentum flux of a plane wave. Explicit fields of the TM and TE modes and the check of Maxwell's equations and boundary conditions are given in Appendix A. Gaussian units and Tamm's notation [4] are used: $T_{ik}=\frac{1}{4\pi}(E_iE_k+H_iH_k)-\frac{\delta_{ik}}{8\pi}(E^2+H^2)$, the force on bodies inside a closed surface is $\oint T_{ik}n_k\,dS$, and the momentum flux is $\Pi_{ik}=-T_{ik}$. We need the component

$$T_{zz}=\frac{1}{8\pi}\left(E_z^2+H_z^2-E_x^2-E_y^2-H_x^2-H_y^2\right),$$

and denote the pressure on a horizontal area element by $p\equiv-\langle T_{zz}\rangle$.

Pressure of a single mode. Average T_{zz} over the cross-section ab and over the period. For $n_x,n_y\geq 1$ every factor $\sin^2$ or $\cos^2$ gives $\frac{1}{2}$, and for a TM mode

$$8\pi\langle T_{zz}\rangle=\frac{A^2}{8}\left[\cos^2 k_z z\left(1-\frac{k^2}{K^2}\right)-\frac{k_z^2}{K^2}\sin^2 k_z z\right]=-\frac{A^2}{8}\frac{k_z^2}{K^2},$$

since $1-k^2/K^2=-k_z^2/K^2$. The energy density of the same mode averaged over the volume is $8\pi\overline{w}=\frac{A^2}{8}\frac{k^2}{K^2}$. Therefore

$$p=-\langle T_{zz}\rangle=\overline{w}\,\frac{k_z^2}{k^2}.$$

For a TE mode the same steps give the same result. If one of the transverse indices is zero (possible only for TE), the overall factor doubles both in the pressure and in the energy, and the ratio does not change; the TM mode with $n_z=0$ has $k_z=0$ and produces no pressure.

Independence of z. The z-dependence has dropped out ($\cos^2+\sin^2=1$). The general reason: in vacuum $\partial T_{zk}/\partial x_k=\frac{1}{4\pi c}\partial_t[\mathbf{E}\times\mathbf{H}]_z$, and the period average of a time derivative vanishes. On integrating over the cross-section, fluxes through the side walls $T_{zx}=\frac{1}{4\pi}(E_zE_x+H_zH_x)$ remain; on the walls $x=0,a$ $E_z=0$ (tangential field) and $H_x=0$ (normal field), i.e. $T_{zx}=0$, and likewise $T_{zy}=0$ on $y=0,b$. Hence $\frac{d}{dz}\int\langle T_{zz}\rangle\,dx\,dy=0$. Note that only the cross-section integral is independent of z; at an individual point (x,y) the value of T_{zz} varies with z, but it is the integral that enters the force. This fact will be used in the next section.

Sum over zero-point oscillations. Each mode is an oscillator with energy $\hbar\omega/2$ in the ground state, shared equally between the electric and magnetic parts — the same distribution that the classical period average gives if one sets $\overline{w}V=\hbar\omega/2$. Therefore the pressure in a cavity is

$$p=\frac{\hbar c}{2V}\sum_{\text{modes}}\frac{k_z^2}{k}.$$

It differs from the energy sum $E=\frac{\hbar c}{2}\sum k$ only by the summand: k_z^2/k instead of k.

Identity with the Hamiltonian approach. With $k_z=\pi n_z/h$ we have $\partial k/\partial h=-k_z^2/(kh)$, and for every mode

$$-\frac{\partial}{\partial h}\frac{\hbar ck}{2}=\frac{\hbar c}{2h}\frac{k_z^2}{k}=S\,p_{\text{mode}},\qquad V=Sh.$$

The force found from the stress tensor coincides with $-\partial E/\partial h$ mode by mode, even before any regularization (the Hellmann–Feynman theorem [5]; in other words, conservation of the adiabatic invariant E/ω of each mode under a slow displacement of the wall). The Hamiltonian force of the previous section is thereby confirmed by an independent method.

Correction of Appendix B of [1]. There the pressure on the xy face was computed as $\langle H_x^2+H_y^2\rangle/8\pi$. For TM modes this omits the term $-\langle E_z^2\rangle/8\pi$: with it the ratio of pressure to energy density equals k_z^2/k^2 for modes of both types, not 1 for the electric mode (for the mode $(2,1,3)$ with $a:b:h=1:1.3:2.7$ it is 0.2119 instead of 1). Consequently the value 1 for the electric mode and the inequality $F/S\geq\delta(E/V)$ of [1] that rests on it are incorrect. The Hamiltonian derivation of Appendix C is not affected: by the identity just proved it coincides with the stress-tensor one mode by mode.

Regularization. With a cutoff $f(k/k_m)$ the derivative of the mode energy is $-\partial_h\left[\frac{\hbar ck}{2}f\right]=\frac{\hbar c}{2h}\frac{k_z^2}{k}\left[f+\frac{k}{k_m}f'\right]$, while the regularized stress-tensor sum $\sum(k_z^2/k)f$ does not contain the

second term. At finite ak_m the two expressions therefore differ, but as $ak_m \to \infty$ the difference vanishes. For the honeycomb plate this is verified numerically (section "Checks and estimates").

Symmetry check. The renormalized electromagnetic tensor $\langle T_{\mu\nu}\rangle$ is traceless. Between two parallel plates the boundary conditions are invariant under boosts along the plates; together with the trace condition this gives $\langle \Pi_{zz}\rangle = 3\langle u\rangle$ (Brown and Maclay [6]). A honeycomb cell is a waveguide along z: far from its ends its boundary conditions are invariant under boosts along z, so in the (t,z) block $\langle T_{\mu\nu}\rangle \propto \eta_{\mu\nu}$ and $\langle \Pi_{zz}\rangle = -\langle u\rangle$, i.e. $|F/S| = \delta(E/V)$ — the same value as the Hamiltonian approach gives. The argument is exact for a perfect conductor in the limit $ak_m \to \infty$ and up to end corrections of order a/l.

WHERE THE FORCE ACTS

Jump at the metal surface. At the face $z = 0$ (vacuum outside, inside the metal the fields vanish) on the vacuum side $E_x = E_y = 0$, $H_z = 0$, while the normal $\mathbf{E}$ and the tangential $\mathbf{H}$ jump by $4\pi\sigma$ and $\frac{4\pi}{c}\mathbf{i}$. Therefore

$$T_{zz}\big|_{\text{vacuum}} = \frac{1}{8\pi}\left(E_z^2 - H_x^2 - H_y^2\right), \qquad T_{zz}\big|_{\text{metal}} = 0.$$

For a TM mode at $z = 0$ $\langle E_z^2\rangle = \frac{A^2}{8}$, $\langle H_x^2 + H_y^2\rangle = \frac{A^2}{8}\frac{k^2}{K^2}$, whence $-\langle T_{zz}\rangle(0^+) = \overline{w}k_z^2/k^2$ — the same value as in the bulk of the cavity; for a TE mode $E_z = 0$ and the result is the same. The jump $[\![T_{zz}]\!] = T_{zz}|_{\text{metal}} - T_{zz}|_{\text{vacuum}} = p$ is the surface force per unit area acting on the charges $\sigma = E_n/4\pi$ in the field E_n and on the currents $\mathbf{i} = \frac{c}{4\pi}[\mathbf{n}\times\mathbf{H}]$ in the field $\mathbf{H}_t$:

$$\frac{f_z}{S} = \frac{1}{8\pi}\left(\langle H_t^2\rangle - \langle E_n^2\rangle\right).$$

In the vacuum state $\langle\sigma\rangle = 0$ and $\langle\mathbf{i}\rangle = 0$, but the correlations $\langle\sigma E_n\rangle$ and $\langle\mathbf{i}\times\mathbf{H}\rangle$ are non-zero: the force is quadratic in the fields. For TE modes there is no surface charge on the xy faces at all, and the jump is produced by the currents alone; this is an explicit example of a jump of the tensor at a surface that carries no charge.

The plate has two such internal boundaries: the bottom of the cells and the smooth face. On each the jump equals the pressure of the adjacent cavity, $p_{\text{cell}} = \frac{\hbar c}{2a^2 l}\sum_{\text{cell}}\frac{k_z^2}{k}$ and $p_{\text{big}} = \frac{\hbar c}{2L^2(L-l)}\sum_{\text{big}}\frac{k_z^2}{k}$; inside the metal $T_{ik} = 0$.

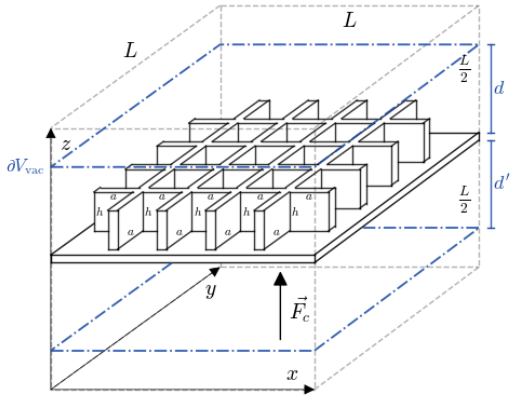


Fig.2. Faces of the pillbox ∂V_{vac} at $z = \pm d$.

The pillbox. Enclose the plate by a closed surface ∂V_{vac} with faces $z = +d$ and $z = -d'$ ($0 < d \le l$, $0 < d' \le L - l$) spanning the whole cross-section $L \times L$ (Fig. 2); the side faces lie on the cube walls, where $T_{zx} = T_{zy} = 0$. Then

$$F_z = \oint_{\partial V_{\text{vac}}} \langle T_{zz}\rangle\, da = \left[\langle T_{zz}\rangle(d) - \langle T_{zz}\rangle(-d')\right] L^2 = (p_{\text{big}} - p_{\text{cell}})L^2.$$

By the proven independence of z the result does not depend on d, d'. For $d, d' \to 0$ it is the sum of the two jumps at the plate faces — the force acts on the metal surfaces; for $d = l$, $d' = L - l$ the faces approach the auxiliary walls of the cube (for $l = L/2$, the faces $z = \pm L/2$), understood as a limit from inside the cavities. On the honeycomb side the face crosses the cell walls; for perfectly conducting walls of zero thickness they do not contribute to the integral. $F_z > 0$ (a push towards the cells) when $p_{\text{big}} > p_{\text{cell}}$.

The flux through ∂V_{vac} is the momentum flux of the field through a purely vacuum boundary, not a force applied directly to matter. It is non-zero only because the mode spectrum, and hence $\langle T_{zz}\rangle$, differs on the honeycomb side and on the smooth side.

WHY THE STANDARD MOMENTUM BALANCE DOES NOT SEE THIS

The standard momentum balance equation of the electrodynamics of continuous media (Tamm, § 105 [4]) reads

$$\nabla\cdot\mathbf{T} = \frac{\partial\mathbf{g}}{\partial t} + \mathbf{f}_{\text{vol}}, \qquad \mathbf{g} = \frac{1}{c^2}\mathbf{S} = \frac{1}{4\pi c}[\mathbf{E}\times\mathbf{H}],$$

where $\mathbf{g}$ is the field momentum density and $\mathbf{f}_{\text{vol}}$ the density of the volume ponderomotive force. For vacuum fluctuations of a static honeycomb system $\langle\mathbf{S}\rangle = 0$, hence $\langle\partial\mathbf{g}/\partial t\rangle = 0$. If the field momentum is identified with $\mathbf{g}$, then with $\langle\mathbf{S}\rangle = 0$ there is no momentum transport either — and a one-body force seems impossible.

The transition from volume forces to stresses relies on Gauss's theorem, which Tamm applies (§ 33, Eq. 33.7) under the explicit condition that the tensor components be continuous inside the volume; surfaces of discontinuity are excluded there, and later (§ 34, Eq. 34.8) only their surface charges are considered. But, as shown above,

a vacuum–metal boundary may carry no surface charge and still produce a finite jump of the stress tensor. This case is considered neither in § 33 nor anywhere else in Tamm's book, including § 105. For a volume V containing internal surfaces of discontinuity S_j, Gauss's theorem must be written as

$$\int_V \frac{\partial T_{ik}}{\partial x_k}\, dV = \oint_{\partial V_{\rm ext}} T_{ik}\, n_k\, dS - \sum_j \int_{S_j} [\![T_{ik}]\!]\, n_k\, dS_j,$$

where $[\![\cdot]\!]$ is the jump of a quantity across S_j. The customary identification of the field momentum with $\mathbf{g} = \mathbf{S}/c^2$ keeps only the outer surface integral and loses the jump terms. For the honeycomb plate it is precisely these terms that make up the whole force: the sum of the two jumps at the plate faces equals the flux through the pillbox.

Two remarks make the interpretation unambiguous. First, as already stated, it is the momentum of the field (and of the plate absorbing it) that is not conserved, not the total momentum of a closed laboratory system. Second, the renormalized vacuum stress $\langle T_{\mu\nu} \rangle$ is a physically measurable quantity — it is the Casimir effect itself — so the jump terms do not require the assumption of a material ether: their value is fixed by the same $\langle T_{\mu\nu} \rangle$ that has been confirmed experimentally for the symmetric configuration.

CHECKS AND ESTIMATES

Two Casimir plates. Plates at $z = 0$ and $z = a$, outer cavities up to $z = \pm L/2$. In the gap $k_z = \pi n/a$ and the transverse spectrum is continuous; per unit area (two polarizations)

$$p_{\rm in} = \frac{\hbar c}{a} \sum_{n \geq 1} \int \frac{d^2 k_\parallel}{(2\pi)^2}\, \frac{k_z^2}{k},$$

and outside the sum over n is replaced by $\frac{a}{\pi} \int dk_z$. By the independence of z the value of $p_{\rm in}$ is the same in the middle of the gap and at the inner surface of each plate, and $p_{\rm out}$ is the same at the outer surface and at the faces $z = \pm L/2$. With the cutoff e^{-k/k_m} the difference $p_{\rm in} - p_{\rm out}$ equals -0.041043, -0.041110, -0.041120 in units of $\hbar c/a^4$ for $k_m a = 20$, 50, 100 and tends to $-\pi^2/240 = -0.041123$. The sign means attraction; with the Casimir energy density $u = -\pi^2 \hbar c/720 a^4$ this is the relation $p = 3u$ [6].

Stress-tensor sum for the honeycomb. For the honeycomb plate the stress-tensor sum $\sum k_z^2/k$ with the regulator $f = 1/(1 + k^4/k_m^4)$ was computed by the same two-dimensional summation as the energy sum in [1]. Denote by D_E and D_T the energy and stress-tensor "sum minus integral" differences. At small $a k_m$ they differ even in sign (the f' term), but at $a k_m = 9$ and 14 we obtain $D_T + D_E = 1 \cdot 10^{-4}$ and $4 \cdot 10^{-6}$, respectively, with $|D_E| \approx 0.012$: in the limit $a k_m \to \infty$ the stress-tensor and Hamiltonian derivations give the same force, $|F/S| = \delta(E/V)$, in agreement with the boost argument.

Thrust estimate. With $\hbar c\, \pi/a^4 = 0.9932/a_\mu^4$ dyn/cm^2 (a_μ is the cell side in microns) the value $R = 0.0138$ from Table 1 of [1] gives, for gold honeycomb with $a \approx 2$ μm, $F/S \approx 8.55$ dyn/m^2 $= 8.55 \times 10^{-5}$ N/m^2. However, in [1] the series over shells j was stopped as soon as the current term fell below a threshold ε (columns ε and j_{max} of Table 1). The tail terms are positive and decay as j^{-3}, so the discarded remainder $\varepsilon\, j_{max}^3\, \zeta(3, j_{max} + 1) \approx \varepsilon\, j_{max}/2$ is negligible at small $a k_m$ but reaches 15% at $a k_m = 9$ ($j_{max} = 99$). With this correction Table 1 agrees with an independent recalculation (closed form of the u integral, extrapolation in the size of the summation domain) to within 0.13% in every row. The corrected limit $|R| \approx 0.0121$ corresponds to $F/S \approx 7.5$ dyn/m^2. The order of magnitude and the conclusion about practical applicability to satellite orbit correction are unchanged.

LIMITATIONS

(a) *Cell height.* The calculation assumes that the cell walls reach the cube face and the cell is a closed resonator. For real cells of height $h \ll L$ the cell is open into the large cavity, the modes do not separate, and this calculation does not describe such a case.

(b) *Perfect conductor.* The jump is a property of a perfect conductor. In a real metal T_{zz} decays to zero over the skin depth, and the integral of the force over the layer equals the jump as long as the current in the metal is consistent with the field: $\mathbf{j} = \frac{c}{4\pi} \nabla \times \mathbf{H}$.

(c) *Wall thickness.* Cell walls of finite thickness carry no z-force ($T_{zx} = 0$ on them), but they reduce the fraction of the plate area occupied by the cavities.

(d) *Regularization.* Which of the two regularized quantities — energy or stress-tensor — is physical at finite $a k_m$ cannot be decided within the model of a perfect conductor with a mode cutoff; this requires a real $\varepsilon(\omega)$, as in the method of Dzyaloshinskii, Lifshitz and Pitaevskii [8], where the stress in a vacuum gap is found from the Green's function of the field. For a honeycomb cell this requires the Green's function of a rectangular waveguide; this is left for future work.

(e) *Quantum average.* The equality of the electric and magnetic parts of the mode energy in the ground state is used as a standard result for the oscillator and is not derived here.

CONCLUSION

The Casimir honeycomb drive is an example of a physical system in which Noether's theorem directly explains the origin of a macroscopic force: **the absence of spatial homogeneity leads to non-conservation of the field momentum**, and hence to thrust. An independent

calculation through the Maxwell stress tensor built from the mode fields reproduces the same force — identically mode by mode and, after regularization, in the limit $ak_m \to \infty$ — and corrects an error in Appendix B of [1]. The stress-tensor calculation shows that the force acts on the metal surfaces as a jump of $\langle T_{zz}\rangle$, while the standard momentum balance of Tamm's § 105, reduced to $\mathbf{g} = \mathbf{S}/c^2$, does not contain these jump terms. The total momentum of the system plate + field + support is conserved. An experimental test of the effect could confirm the link between the geometry of boundary conditions, symmetry and the dynamics of the quantum vacuum.

APPENDIX A. MODE FIELDS

TM-mode fields ($H_z = 0$, $n_x, n_y \geq 1$, $n_z \geq 0$):

$$E_z = A\sin k_x x \sin k_y y \cos k_z z \cos\omega t, \qquad H_z = 0,$$
$$E_x = -A\frac{k_x k_z}{K^2}\cos k_x x \sin k_y y \sin k_z z \cos\omega t, \qquad H_x = -A\frac{k\,k_y}{K^2}\sin k_x x \cos k_y y \cos k_z z \sin\omega t,$$
$$E_y = -A\frac{k_y k_z}{K^2}\sin k_x x \cos k_y y \sin k_z z \cos\omega t, \qquad H_y = A\frac{k\,k_x}{K^2}\cos k_x x \sin k_y y \cos k_z z \sin\omega t.$$

TE-mode fields ($E_z = 0$, $n_z \geq 1$, $(n_x, n_y) \neq (0,0)$):

$$H_z = A\cos k_x x \cos k_y y \sin k_z z \cos\omega t, \qquad E_z = 0,$$
$$H_x = -A\frac{k_x k_z}{K^2}\sin k_x x \cos k_y y \cos k_z z \cos\omega t, \qquad E_x = -A\frac{k\,k_y}{K^2}\cos k_x x \sin k_y y \sin k_z z \sin\omega t,$$
$$H_y = -A\frac{k_y k_z}{K^2}\cos k_x x \sin k_y y \cos k_z z \cos\omega t, \qquad E_y = A\frac{k\,k_x}{K^2}\sin k_x x \cos k_y y \sin k_z z \sin\omega t.$$

Substitution verifies that $\nabla\cdot\mathbf{E} = \nabla\cdot\mathbf{H} = 0$, $\nabla\times\mathbf{E} = -\frac{1}{c}\partial_t\mathbf{H}$, $\nabla\times\mathbf{H} = \frac{1}{c}\partial_t\mathbf{E}$ for $\omega = ck$, and that the perfect-conductor boundary conditions hold: tangential $\mathbf{E}$ and normal $\mathbf{H}$ vanish on all six faces (on $x = 0, a$: $E_y = E_z = 0$, $H_x = 0$; on $y = 0, b$: $E_x = E_z = 0$, $H_y = 0$; on $z = 0, h$: $E_x = E_y = 0$, $H_z = 0$). These relations are verified symbolically and numerically for the mode $(2,1,3)$ with $a : b : h = 1 : 1.3 : 2.7$.

For a TE mode, averaging over the cross-section and the period gives $\langle H_z^2\rangle = \frac{A^2}{8}\sin^2 k_z z$, $\langle H_x^2 + H_y^2\rangle = \frac{A^2}{8}\frac{k_z^2}{K^2}\cos^2 k_z z$, $\langle E_x^2 + E_y^2\rangle = \frac{A^2}{8}\frac{k^2}{K^2}\sin^2 k_z z$, whence $8\pi\langle T_{zz}\rangle = -\frac{A^2}{8}\frac{k_z^2}{K^2}$ and $p = \overline{w}\,k_z^2/k^2$.

ТЕОРЕМА НЕТЕР ТА СТИЛЬНИКОВИЙ РУШІЙ КАЗИМИРА: ПРО НЕЗБЕРЕЖЕННЯ ІМПУЛЬСУ В АСИМЕТРИЧНИХ ВАКУУМНИХ СИСТЕМАХ

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Стільниковий рушій Казимира [1] — ідеально провідна пластина з нанорозмірною стільниковою структурою з одного боку — розглядається з погляду симетрій. Через асиметрію вакуумних мод з двох боків пластини система не є інваріантною відносно трансляцій уздовж нормалі: вакуумний гамільтоніан явно залежить від координати l, і за теоремою Нетер імпульс поля вздовж цієї осі не зберігається. Сила тяги дорівнює $\partial\langle H\rangle/\partial l$. Потім ту саму силу обчислено незалежно — через тензор натягів Максвелла, побудований з полів власних мод резонатора. Помодово обидва виведення збігаються тотожно (теорема Гельмана–Фейнмана), а після регуляризації — у границі $ak_m \to \infty$. Попутно знайдено й виправлено помилку в додатку B роботи [1]: для TM-мод відношення тиску до густини енергії дорівнює k_z^2/k^2, а не одиниці. Тензорний розрахунок показує, де прикладена сила: це стрибок $\langle T_{zz}\rangle$ на поверхні металу, і тому потік імпульсу через будь-яку замкнену вакуумну поверхню, що охоплює пластину, не залежить від положення її граней. Нарешті, показано, що стандартний баланс імпульсу (§ 105 Тамма) при $\langle \mathbf{S}\rangle = 0$ втрачає саме ці стрибкові члени. Не зберігається імпульс поля (і пластини); повний імпульс системи пластина + поле + опора зберігається.

Ключові слова: теорема Нетер, ефект Казимира, квантовий вакуум, незбереження імпульсу, тензор натягів Максвелла, наностільники, тяга Казимира

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